Capital Budgeting for a Tech Infrastructure Investment

Feasibility Analysis of **PayEase** – A Mobile Payment Application

MGTE 34213: MANAGERIAL FINANCE

GROUP ASSIGNMENT

Made with GAMMA

Company & Project Overview

About DigiFlow (Pvt.) Ltd.

A Sri Lankan tech startup founded in 2022, specialising in digital financial solutions. Currently operates two SaaS products serving SME inventory management.

Introducing PayEase

A cross-platform mobile payment app targeting Sri Lankan SMEs. Key features include **QR-code payments**, **LankaPay API integration**, and **invoice generation** — monetised via a freemium model combining premium subscriptions and in-app advertising.

Project Horizon: 5 years (2026–2031) | Year 0 = development & pre-launch

Made with GAMMA

Key Assumptions & Their Basis

Discount Rate — 12%

Benchmarked against CBSL lending rates plus a startup risk premium *(Central Bank of Sri Lanka, Annual Report 2024)*

Revenue Growth — Declining Curve

35% (Y1–Y2) → 30% (Y3) → 25% (Y4) → 20% (Y5), reflecting the standard app maturity curve *(Ross et al., 2019)*

Operating Costs & Tax

Operating cost growth: **10% p.a.** | Corporate income tax: **30%** on positive EBIT only | Straight-line depreciation: **Rs. 440,000/year** over 5 years

Financing Structure

Fully equity-funded — no debt financing. LankaPay API assumptions sourced from *LankaPay Annual Review 2023/2024*

Made with GAMMA

Initial Investment & Cost Breakdown

Total Initial Outlay

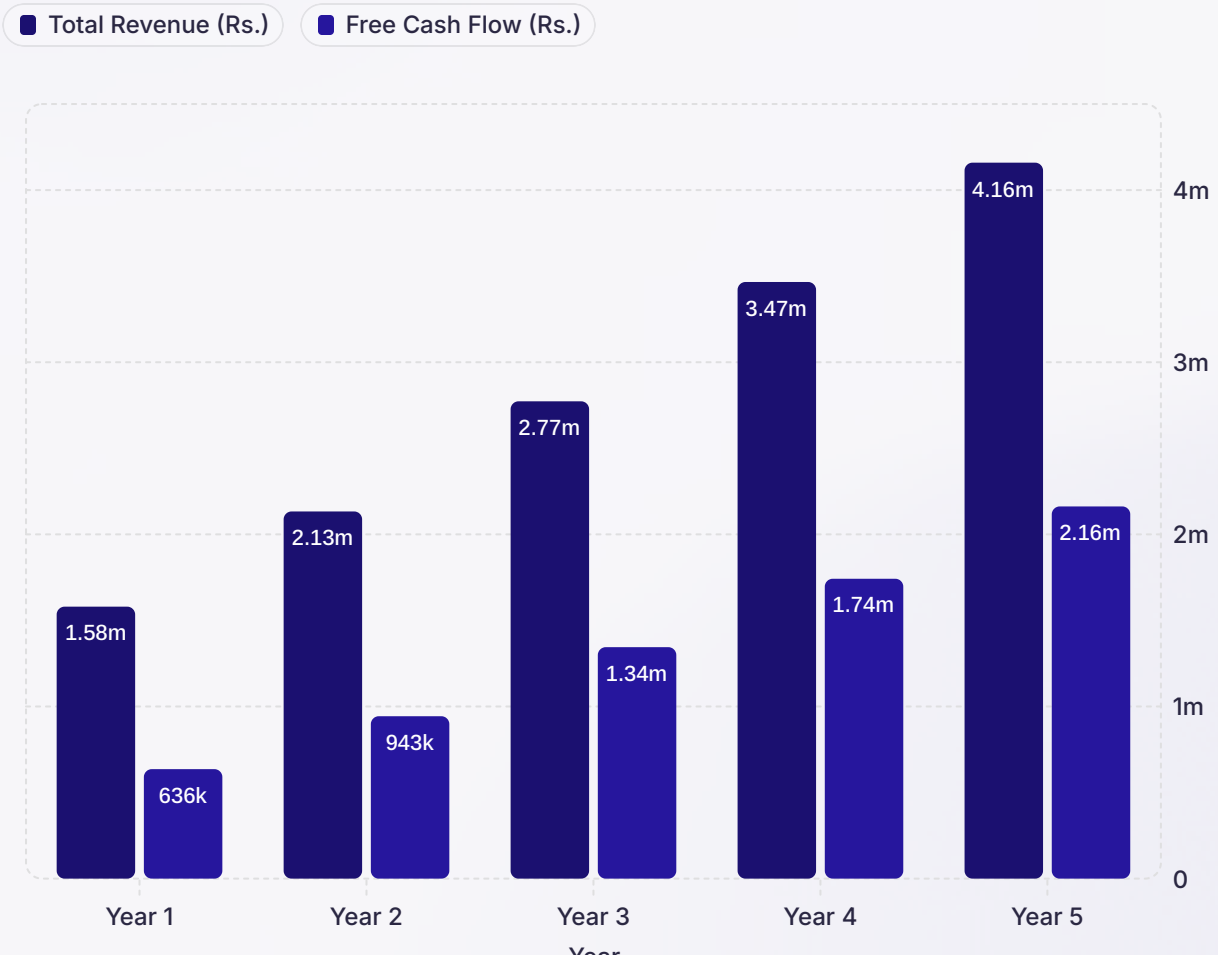
Rs. 2,200,000

Fully equity-funded pre-launch capital expenditure covering all development, infrastructure, and go-to-market costs.

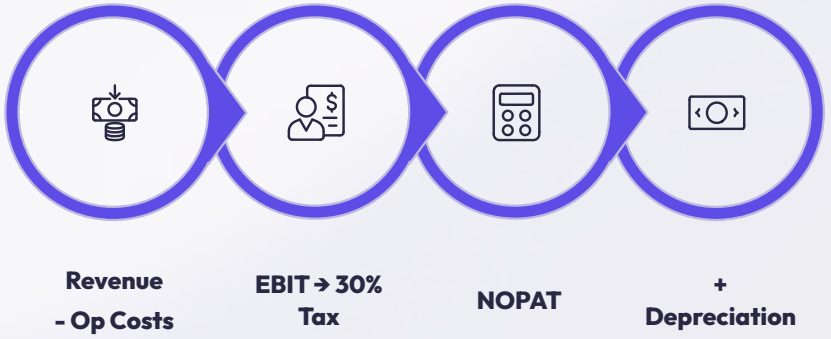
Cost Component	Amount (Rs.)
Software development (4 devs × 6 months)	1,080,000
UX/UI design & prototyping	200,000
QA & third-party testing	150,000
Cloud server setup (AWS)	200,000
API integration (LankaPay & banking)	220,000
Pre-launch marketing & branding	250,000
Contingency (5%)	100,000

Made with GAMMA

Projected Free Cash Flow (Years 1–5)



How FCF is Derived



Depreciation is a non-cash charge — it shields income from tax but is added back when computing actual cash flow *(Brealey, Myers & Allen, 2020)*.

Made with GAMMA

Net Present Value (NPV)

Year	FCF (Rs.)	Discount Factor (12%)	PV (Rs.)
1	636,000	0.8929	567,890
2	943,390	0.7972	752,107
3	1,344,610	0.7118	957,174
4	1,742,000	0.6355	1,107,011
5	2,162,157	0.5674	1,227,847

NPV Result

Rs. +2,436,029

Formula: NPV = Σ [CFt ÷ $(1 + r)^t$] – Initial Outlay

The positive NPV confirms the project generates returns **above the 12% required rate**, creating shareholder value.

DECISION: ACCEPT (NPV > 0)

Made with GAMMA

Internal Rate of Return (IRR)

Interpolation Calculation

Trial Rate	NPV (Rs.)
Low — 40%	+94,967
High — 43%	–42,785

IRR = 40% + [94,967 ÷ (94,967 + 42,785)] × 3%

IRR ≈ 42%

What This Means

The IRR exceeds the cost of capital by **more than 3×**, providing a wide safety margin against model uncertainty and cost overruns.

DECISION: ACCEPT (IRR > 12% WACC)

IRR

42% — the project's internal return rate

WACC

12% — the minimum required rate

Made with GAMMA

Payback Period

Cumulative Cash Flow Recovery

Year	FCF (Rs.)	Cumulative FCF (Rs.)
1	636,000	–1,564,000
2	943,390	–620,610
3	1,344,610	+723,900
4	1,742,000	+2,465,900
5	2,162,157	+4,628,057

Exact Payback Calculation

2 years + (601,100 ÷ 1,344,610) × 12 months

2 Yrs 5 Months

Rapid recovery **well within** the 5-year project life. Minimal downside risk exposure — investors recoup capital early in the project cycle.

DECISION: ACCEPT

Made with GAMMA

Profitability Index (PI)

PI Formula & Result

PI = Total NPV ÷ Initial Investment

PI = Rs. 2,436,029 ÷ Rs. 2,200,000

PI = 1.11

DECISION: ACCEPT (PI > 1.0)

Interpretation & Use Case

Value per Rupee

Every **Rs. 1 invested** generates **Rs. 1.11 in NPV** — confirming efficient use of equity capital.

Capital Rationing Tool

PI is especially valuable when comparing competing projects under budget constraints — it ranks projects by value created per unit of investment *(Ross et al., 2019)*.

Made with GAMMA

Summary of Appraisal Results

All four capital budgeting methods unanimously support acceptance of the PayEase project.

Technique	Result	Benchmark	Decision
Net Present Value (NPV)	**Rs. +2,436,029**	Greater than Rs. 0	Accept
Internal Rate of Return (IRR)	**42%**	Greater than 12% WACC	Accept
Payback Period	**2 years 5 months**	Less than 5 years	Accept
Profitability Index (PI)	**1.11**	Greater than 1.0	Accept

Conclusion: PayEase presents a financially sound investment opportunity for DigiFlow (Pvt.) Ltd. — delivering strong NPV, a high IRR well above the cost of capital, rapid payback, and efficient capital deployment as confirmed by a PI exceeding 1.0.

Made with GAMMA

Conclusion & Recommendation

Conclusion

Rs. 2.2M investment justified across all four appraisal methods

Recommendation

- **DigiFlow (Pvt.) Ltd. should proceed with PayEase**
- Conditions advised:
  - Target 400+ paying subscribers within first 6 months post-launch
  - Review actual vs. projected cash flows at end of Year 2
  - Obtain CBSL regulatory clearance before launch
  - Adopt Agile project management to control timelines and costs

Made with GAMMA

References

- Ross, S. A., Westerfield, R. W., & Jordan, B. D. (2019). *Fundamentals of Corporate Finance* (12th ed.). McGraw-Hill Education.
- Central Bank of Sri Lanka. (2025). *Annual Report 2024.* Colombo: CBSL.
- LankaPay (Pvt.) Ltd. (2024). *Annual Review 2023/2024.* Colombo: LankaPay.
- Brealey, R. A., Myers, S. C., & Allen, F. (2020). *Principles of Corporate Finance* (13th ed.). McGraw-Hill Education.

Made with GAMMA